

# Customer Churn Analysis Project Report

## Project Title

**Customer Churn Analysis and Prediction Using SQL, Python, Power BI, and Machine Learning**

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## 1. Introduction

Customer churn is a major challenge in the telecom industry. When customers stop using a company's services and switch to another provider, it directly affects the company's revenue and customer base. Because of this, telecom companies focus heavily on understanding why customers leave and how they can improve customer retention.

The main purpose of this project is to analyze customer churn patterns using data analytics and machine learning techniques. By studying customer behavior, billing information, internet services, and contract details, the project aims to identify the key factors that influence churn.

This project was carried out using SQL for initial business analysis, Python for data cleaning and exploratory analysis, Machine Learning for churn prediction, Power BI for dashboard development, and Streamlit for deployment.

The overall project demonstrates how data analytics can help businesses make better decisions and reduce customer loss.

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## 2. Objectives of the Project

The main objectives of this project are:

- To understand customer churn behavior in the telecom industry
  - To identify the important factors influencing customer churn
  - To clean and preprocess the dataset for analysis
  - To perform SQL-based business analysis
  - To create visualizations and identify meaningful insights
  - To build a machine learning model for churn prediction
  - To design an interactive Power BI dashboard
  - To deploy the prediction model using Streamlit
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### 3. Tools and Technologies Used

| Tool / Technology | Purpose                                       |
|-------------------|-----------------------------------------------|
| SQL (PostgreSQL)  | Business analysis and querying                |
| Python            | Data cleaning, analysis, and machine learning |
| Pandas            | Data manipulation                             |
| NumPy             | Numerical operations                          |
| Matplotlib        | Data visualization                            |
| Seaborn           | Statistical visualization                     |
| Scikit-learn      | Machine learning                              |
| Power BI          | Dashboard creation and reporting              |
| Streamlit         | Web application deployment                    |
| Jupyter Notebook  | Development environment                       |

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### 4. Dataset Information

The dataset used in this project is a telecom customer churn dataset containing customer demographics, billing details, service usage information, and churn status.

The dataset contains **7043 customer records** and includes both categorical and numerical features.

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#### Original Dataset Columns

| Column Name      | Data Type |
|------------------|-----------|
| customerID       | object    |
| gender           | object    |
| SeniorCitizen    | int64     |
| Partner          | object    |
| Dependents       | object    |
| tenure           | int64     |
| PhoneService     | object    |
| MultipleLines    | object    |
| InternetService  | object    |
| OnlineSecurity   | object    |
| OnlineBackup     | object    |
| DeviceProtection | object    |
| TechSupport      | object    |
| StreamingTV      | object    |
| StreamingMovies  | object    |
| Contract         | object    |
| PaperlessBilling | object    |
| PaymentMethod    | object    |

|                |         |
|----------------|---------|
| MonthlyCharges | float64 |
| TotalCharges   | object  |
| Churn          | object  |

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## Dataset After Cleaning and Feature Selection

| Column Name      | Data Type |
|------------------|-----------|
| gender           | object    |
| Partner          | object    |
| tenure           | int64     |
| PhoneService     | object    |
| InternetService  | object    |
| Contract         | object    |
| PaperlessBilling | object    |
| PaymentMethod    | object    |
| MonthlyCharges   | float64   |
| TotalCharges     | float64   |
| Churn            | int64     |

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## 5. SQL Analysis

The first part of the project started with loading the telecom churn dataset into PostgreSQL. Before moving into Python analysis, SQL was used to understand the dataset structure and generate some basic business statistics.

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### SQL Workflow

The following steps were performed during the SQL analysis phase:

1. Imported the dataset into PostgreSQL
  2. Verified the dataset structure and records
  3. Performed customer-related business analysis using SQL queries
  4. Generated customer statistics and churn-related information
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### SQL Queries Performed

Some of the SQL queries performed in this project include:

- Total number of customers
- Count of churned and retained customers
- Customer distribution by contract type

- Different payment methods used by customers
  - Average monthly charges
  - Maximum and minimum tenure
  - Gender-wise customer count
  - Percentage of churned customers
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## SQL Concepts Used

The following SQL concepts were used:

- Aggregate Functions
  - GROUP BY
  - COUNT
  - AVG
  - Percentage Calculations
  - Filtering and Sorting
- 

## SQL Analysis Summary

The SQL analysis helped in understanding the dataset before moving to advanced analytics in Python. It provided useful information about customer distribution, churn counts, payment methods, and contract types.

This phase also helped verify the dataset quality and business-related statistics.

## Outcomes

| churn         | count         |
|---------------|---------------|
| abc Filter... | abc Filter... |
| No            | 5174          |
| Yes           | 1869          |

| gender        | count         |
|---------------|---------------|
| abc Filter... | abc Filter... |
| Female        | 3488          |
| Male          | 3555          |

## 6. Data Cleaning and Preprocessing

After completing the SQL analysis, the dataset was processed using Python for cleaning and preprocessing.

Data cleaning is an important step because raw datasets often contain missing values, incorrect datatypes, and inconsistent records that can affect the analysis and machine learning model performance.

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### Preprocessing Steps Performed

The following preprocessing tasks were completed:

- Checked for missing values
- Verified duplicate records
- Converted TotalCharges into numeric datatype
- Replaced blank values with NaN
- Verified column datatypes
- Encoded churn values for machine learning
- Removed unnecessary columns
- Prepared features for analysis

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### Data Cleaning Summary

The preprocessing stage improved the quality of the dataset and prepared it for exploratory analysis and model building.

### Outcomes

```
Data columns (total 11 columns):
#      Column      Non-Null Count  Dtype
---  -
0      gender      7043 non-null    object
1      Partner      7043 non-null    object
2      tenure       7043 non-null    int64
3      PhoneService  7043 non-null    object
4      InternetService 7043 non-null    object
5      Contract      7043 non-null    object
6      PaperlessBilling 7043 non-null    object
7      PaymentMethod  7043 non-null    object
8      MonthlyCharges 7043 non-null    float64
9      TotalCharges  7043 non-null    float64
10     Churn         7043 non-null    int64
dtypes: float64(2), int64(2), object(7)
memory usage: 605.4+ KB
```

## 7. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand customer behavior and identify patterns related to customer churn.

Several visualizations were created to compare customer categories and observe how different features influence churn.

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### Areas Analyzed

The following areas were analyzed during EDA:

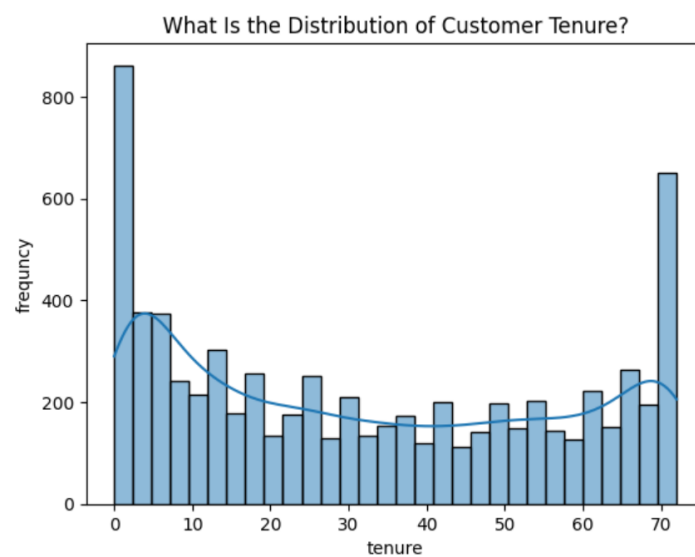
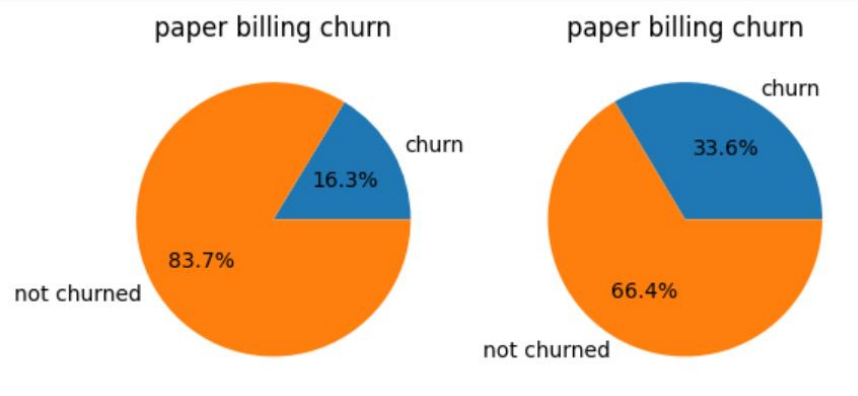
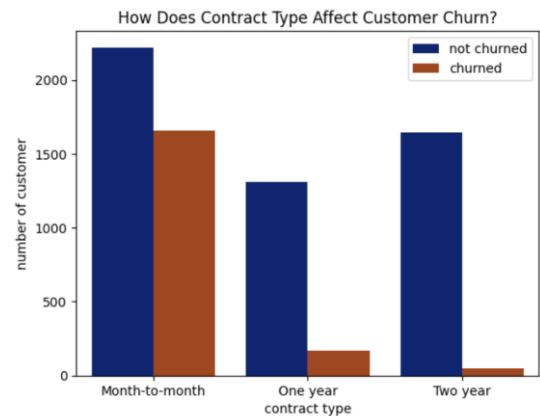
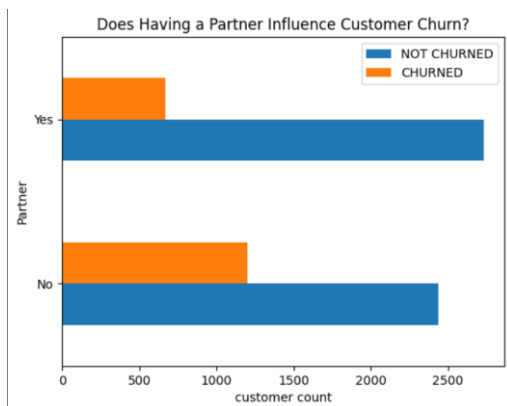
- Gender
  - Partner status
  - Phone service
  - Internet service
  - Contract type
  - Paperless billing
  - Payment methods
  - Customer tenure
  - Monthly charges
  - Total charges
- 

### Visualizations Created

The following visualizations were created:

- Gender vs Churn
- Contract Type vs Churn
- Internet Service vs Churn
- Payment Method vs Churn
- Monthly Charges Distribution
- Tenure Analysis
- Churn Count Plot
- Correlation Heatmap

# Outcomes



## **8. Key Business Insights**

### **1. Month-to-Month Customers Have Higher Churn**

Customers with month-to-month contracts showed a much higher churn rate compared to customers with yearly contracts. This indicates that long-term contracts help improve customer retention.

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### **2. Fiber Optic Customers Show More Churn**

Customers using Fiber Optic internet services had higher churn rates compared to DSL users and customers without internet service.

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### **3. Electronic Check Users Are More Likely to Churn**

Customers using electronic check as their payment method had the highest churn rate. Customers using automatic payment methods such as bank transfer and credit card showed lower churn.

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### **4. Customers With Lower Tenure Tend to Leave More**

Customers who recently joined the company were more likely to churn, while long-term customers were more likely to stay.

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### **5. Higher Monthly Charges Increase Churn Risk**

Customers paying higher monthly charges showed higher churn probability, suggesting possible dissatisfaction with expensive plans.



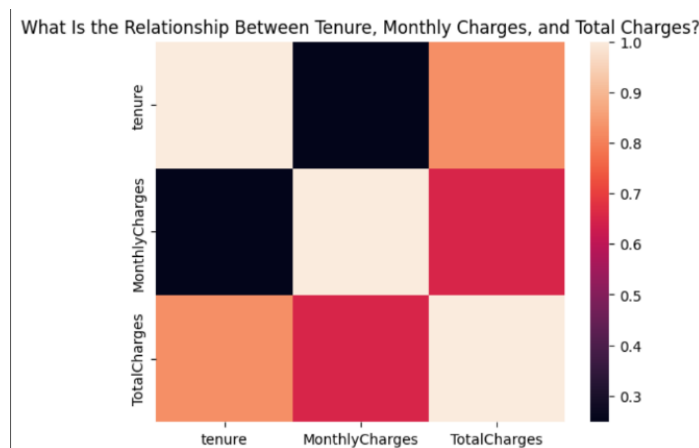
## 9. Correlation Analysis

Correlation analysis was performed between numerical columns such as tenure, MonthlyCharges, and TotalCharges.

A heatmap was created to visualize the relationships between these variables.

The analysis showed that TotalCharges strongly depend on customer tenure, which means long-term customers contribute more overall revenue.

## Outcomes



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## 10. Machine Learning Model

A machine learning model was developed to predict whether a customer is likely to churn.

The goal of the model is to help businesses identify high-risk customers in advance so that retention strategies can be applied.

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## Machine Learning Workflow

The following steps were followed during model development:

1. Data preprocessing
2. Feature transformation

- 3. Train-test split
- 4. Model training
- 5. Model evaluation
- 6. Model saving

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## Algorithms Used

Two machine learning algorithms were tested:

| Algorithm                | Accuracy          |
|--------------------------|-------------------|
| Logistic Regression      | Approximately 81% |
| Decision Tree Classifier | Approximately 79% |

Among the tested models, Logistic Regression performed better and was selected as the final model.

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## Preprocessing Techniques Used

The following preprocessing techniques were applied:

- Pipeline
- StandardScaler
- OneHotEncoder
- ColumnTransformer

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## Evaluation Metrics

The models were evaluated using:

- Accuracy Score
- Precision
- Recall
- F1-Score
- Classification Report

```
classification report
              precision    recall  f1-score   support

     0       0.86      0.90      0.88       1036
     1       0.68      0.58      0.63        373

 accuracy          0.82       1409
 macro avg          0.77       1409
 weighted avg       0.81       1409
```

## Model Saving

The final model was saved using Joblib so that it could be used later in the Streamlit application.

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# 11. Power BI Dashboard Development

An interactive Power BI dashboard was created to visualize churn patterns and support business decision-making.

The dashboard was designed to provide both high-level KPIs and detailed churn insights.

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## Power BI Data Preparation

Power Query was used for:

- Data cleaning
- Data transformation
- Formatting
- Preparing data for visualization

DAX measures were also created for KPI calculations.

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## Dashboard Structure

The dashboard contains:

- 1 Overview Page
- 5 Insight Pages

Each page focuses on different churn-related analyses.

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## KPI Cards Included

The dashboard includes the following KPI cards:

- Total Customers

- Churn Rate
- Churned Customers
- Retained Customers
- Total Charges

## Visualizations Used

The following charts were used:

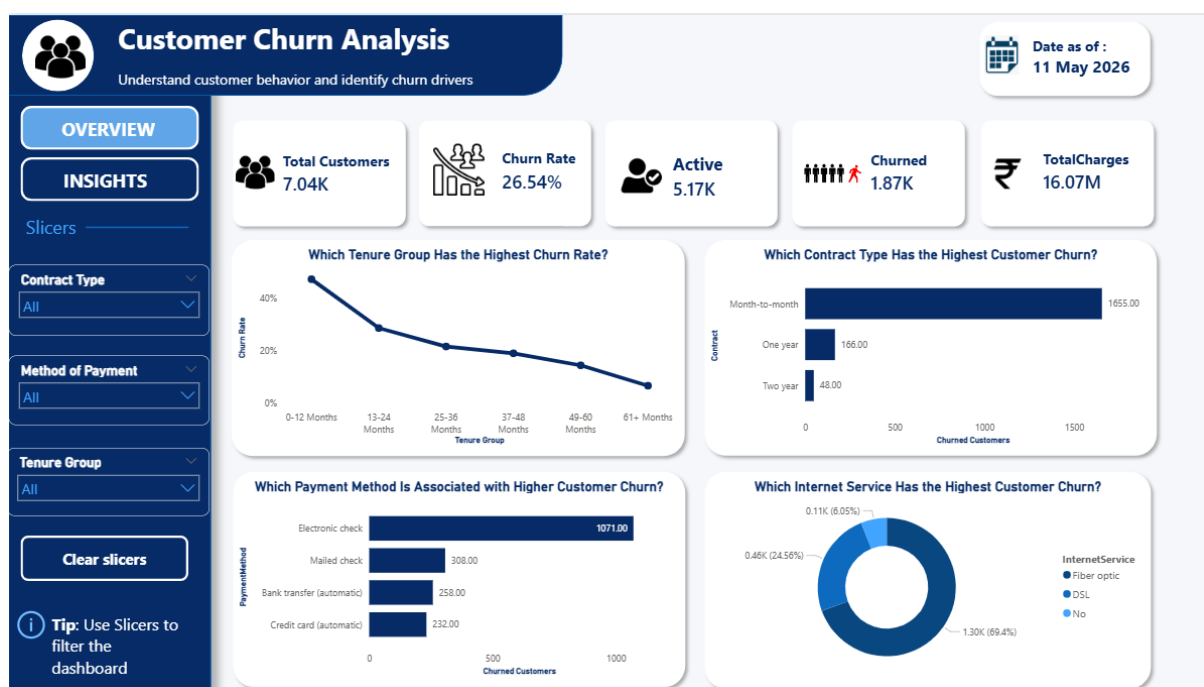
- Bar Charts
- Donut Charts
- Clustered Column Charts
- Line Charts
- KPI Cards
- Heatmaps

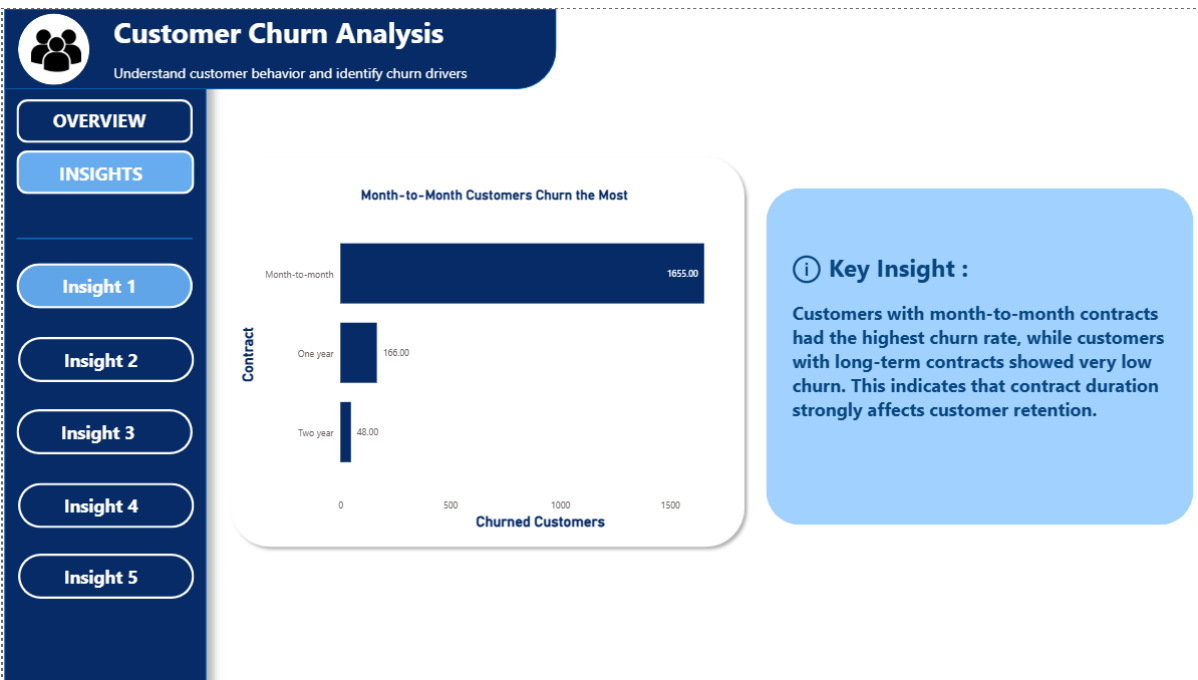
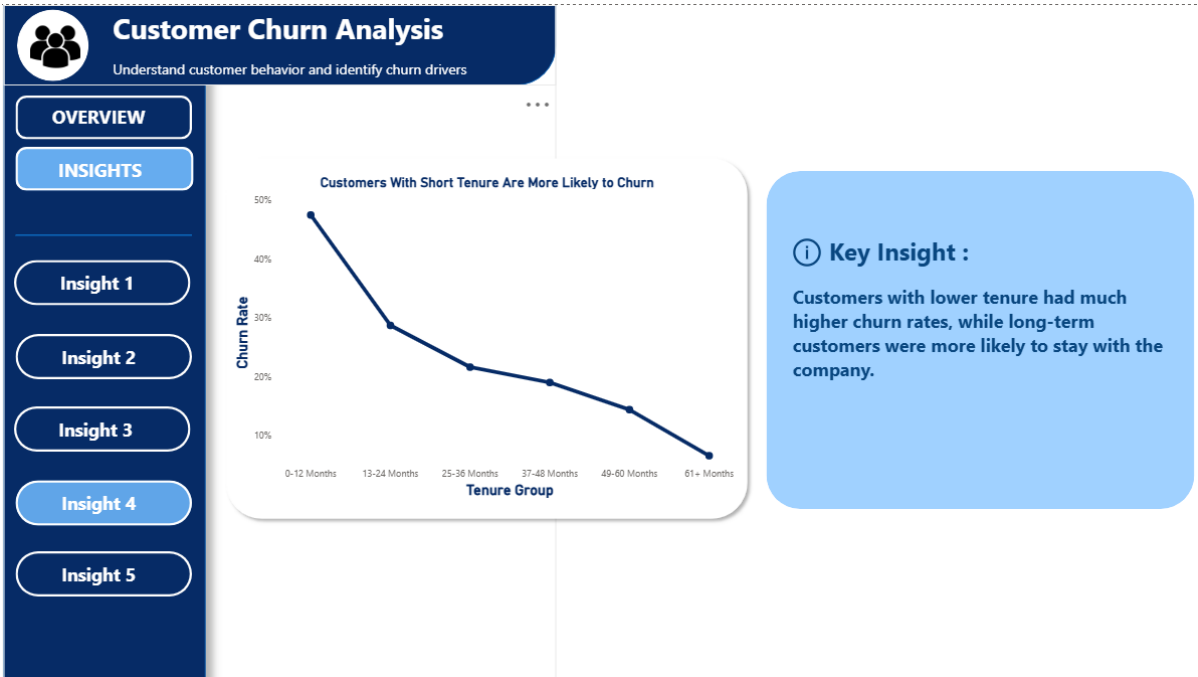
## Interactive Features

The dashboard includes:

- Navigation buttons
- Interactive slicers
- Dynamic filtering
- Cross-filtering between visuals

## Outcomes





# 12. Streamlit Application

A Streamlit web application was developed to deploy the machine learning model in a user-friendly way.

The application allows users to enter customer information and instantly predict whether the customer is likely to churn.

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## Features of the Application

- Interactive user interface
  - Real-time churn prediction
  - Business-themed design
  - Churn probability display
  - Recommended business actions
- 

## User Inputs

The application accepts customer details such as:

- Gender
  - Partner status
  - Tenure
  - Internet service
  - Contract type
  - Payment method
  - Monthly charges
- 

## Automatic Calculation

The TotalCharges value is automatically calculated using:

$$\text{TotalCharges} = \text{tenure} \times \text{MonthlyCharges}$$

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## Prediction Workflow

1. User enters customer details
2. Input data is converted into a dataframe

3. Saved machine learning model is loaded
4. Prediction is generated
5. Churn probability is displayed

## Outcomes

The screenshot shows a web application titled "Telecom Customer Churn Prediction" running on localhost:8501. The interface is divided into several sections:

- Customer Information:** A form with fields for Name, Age, Gender, Tenure, Monthly Charges, and Total Charges. Each field has a dropdown menu.
- Billing & Contract:** A form with fields for Contract Type, Payment Method, and Payment Status. Each field has a dropdown menu.
- Prediction Result:** A section showing the churn probability as 38.17%. Below this, a green bar indicates the prediction result: "Low Risk: Customer is not likely to churn".
- Recommended Business Actions:** A list of three actions: "Monitor customer satisfaction", "Offer long-term contracts", and "Provide personalized services".

The interface is dark-themed with blue and green accents. A "Chat" button is visible in the top right corner.

## 13. Business Recommendations

Based on the analysis, the following recommendations are suggested:

### 1. Promote Long-Term Contracts

Providing discounts and benefits for yearly contracts may help reduce churn.

### 2. Improve Fiber Optic Service Quality

The company should investigate service-related issues affecting Fiber Optic customers.

### 3. Encourage Automatic Payments

Customers using automatic payment methods showed lower churn rates, so promoting these payment options may improve retention.

## 4. Focus on New Customers

Retention campaigns should target customers with lower tenure because they are more likely to churn.

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## 5. Review Expensive Plans

The company should evaluate pricing strategies and improve the value offered in higher-priced plans.

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# 14. Conclusion

This project successfully analyzed telecom customer churn using SQL, Python, Power BI, and Machine Learning.

The project helped identify important factors affecting customer churn, including:

- Contract type
- Internet service
- Payment method
- Customer tenure
- Monthly charges

The machine learning model successfully predicted churn with good accuracy, and the Power BI dashboard provided interactive business insights.

Overall, this project improved understanding of:

- SQL querying
  - Data preprocessing
  - Exploratory Data Analysis
  - Business intelligence
  - Dashboard development
  - Machine learning
  - Customer behavior analysis
- 

# 15. Future Improvements

Future improvements for this project may include:

- Advanced machine learning models



- Hyperparameter tuning
  - Deep learning approaches
  - Real-time churn prediction
  - Cloud deployment
  - Customer segmentation using clustering
  - Advanced dashboard analytics
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## 16. Skills Demonstrated

This project demonstrates skills in:

- SQL
  - Python Programming
  - Data Cleaning
  - Data Analysis
  - Data Visualization
  - Power BI Dashboarding
  - Machine Learning
  - Business Analytics
  - Predictive Modeling
  - Problem Solving
- 

## 17. Final Outcome

This project provides a complete end-to-end customer churn analysis solution that combines:

- SQL-based business analysis
- Python-based analytics
- Machine learning prediction
- Power BI dashboarding
- Streamlit deployment

The project demonstrates how data analytics and machine learning can help telecom companies improve customer retention, reduce churn, and support better business decisions.